

Advanced Lesson 5 - Exercises: Lists

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MATH 7013 - Advanced Mathematical Modeling

Every list produced below is named, so that later exercises can refer to the list built in an earlier exercise instead of rebuilding it.

```
Clear[a, x, i, j, list3, list4, mat5, courseNumbers, courseNames, me];
```

1: A list holding a name (string), an age (integer) and a major (symbols)

```
me = {"Mikey Ferguson", 27, mathematics}
```

The string carries quotation marks, the age is an exact integer, and "mathematics" is an undefined symbol, so it simply stays as itself.

2: A list of lists - course numbers (integers) and course names (symbols)

```
courseNumbers = {7013, 7023, 7113};  
courseNames = {advancedMathematicalModeling, realAnalysis, numericalAnalysis};  
TableForm[{courseNumbers, courseNames}]
```

3: The integers from 0 to 20, via Table

```
list3 = Table[i, {i, 0, 20}]
```

4: The even integers from 0 to 20, via Table

```
list4 = Table[i, {i, 0, 20, 2}]
```

5: A 4x5 matrix with entries Sin[i] Cos[j], displayed with MatrixForm

```
mat5 = Table[Sin[i] Cos[j], {i, 1, 4}, {j, 1, 5}];  
MatrixForm[mat5]
```

6: Sorting the course numbers and the course names

```
{Sort[courseNumbers], Sort[courseNames]}
```

Sort puts integers in numerical order and symbols in alphabetical order, which is what "canonical order" means for each of those two types.

7: Reverse of the list from Exercise 3

```
Reverse[list3]
```

8: Drop the first three, then the last three, elements of the Exercise 3 list

```
{Drop[list3, 3], Drop[list3, -3]}
```

9: Take the first three, the last three, and elements 3, 4 and 11

```
{Take[list3, 3], Take[list3, -3], list3[[3, 4, 11]]}
```

Take handles the two contiguous cases. Elements 3, 4 and 11 are not contiguous, so Take cannot express them; Part with a list of positions can.

10: Insert "Fun and games" as the second element of the course-name list

```
Insert[courseNames, "Fun and games", 2]
```

11: Replace the second course name with "Fun and games"

```
ReplacePart[courseNames, 2 → "Fun and games"]
```

Insert lengthens the list to four entries; ReplacePart leaves it at three and overwrites one of them.

12: Length of the Exercise 4 list

```
Length[list4]
```

13: Dimensions of the Exercise 5 list of lists

```
Dimensions[mat5]
```

14: Intersection of the Exercise 3 and Exercise 4 lists, in ColumnForm

```
ColumnForm[list3 ∩ list4]
```

Every even integer from 0 to 20 already appears in the list 0 through 20, so the intersection is exactly the Exercise 4 list.

15: Join of the Exercise 3 and Exercise 4 lists, in ColumnForm

```
ColumnForm[Join[list3, list4]]
```

Join concatenates and does not remove duplicates: $21 + 11 = 32$ entries.

16: Partition the Exercise 3 list into sublists of length 4

```
list16 = Partition[list3, 4];
```

```
TableForm[list16]
```

21 elements do not divide evenly into blocks of 4, so Partition discards the leftover element and returns five sublists.

17: Partition into three sublists, each holding sublists of length 2

```
MatrixForm[Partition[Partition[list3, 2], 3]]
```

18: Part extraction - third element of Exercise 3, second element of Exercise 16

```
{list3[[3]], list16[[2]]}
```

19: The entry in row 2, column 3 of the Exercise 5 matrix

```
mat5[[2, 3]]
```

20: The second row of the Exercise 5 matrix

```
mat5[[2]]
```

21: The second column of the Exercise 5 matrix, using Part alone

```
mat5[[All, 2]]
```

All in the first slot means "every row", so the result is the column of second entries. This is Part doing the work that Map would otherwise do.

22: Exp applied directly to a mixed list

```
e^{1,2,a,x,"string",\pi}
```

Power has the Listable attribute, so it threads over the list without any help. Even the string is raised to a power symbolically - Mathematica does not object, it simply cannot simplify it.

23: Log over the same list via Map, then N applied to the result

```
logs = Log /@ {1, 2, a, x, "string",  $\pi$ }
```

```
N[logs]
```

Map is the indirect route: it applies Log to each element separately rather than relying on listability. N then converts whatever is numeric.

24: Apply Plus to the Exercise 4 list

```
Plus @@ list4
```

Apply replaces the head List with Plus, turning {0, 2, ..., 20} into $0 + 2 + \dots + 20 = 110$.

25: Flatten the Exercise 5 matrix

```
Flatten[mat5]
```

The 4 x 5 nested structure collapses into a single list of 20 entries.